TRUNCATION OF THE PUSHFORWARD TO AN OPEN VERSUS VANISHING OF HOM FROM THE KOSZUL COMPLEX, SECOND VARIANT

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The Koszul complexes fit into distinguished triangles under multiplication of the exponents, by the More on Algebra Koszul multiplicativity lemma. invoked at proof lines 10653-10656. Transfers the Hom vanishing from the Koszul complex K to all its powers K_e.
- For an object supported on the closed complement of U, its cohomology classes on X are represented by maps from the powers of the Koszul complex, by the represent-cohomology-class lemma. invoked at proof lines 10659-10663. Converts the Hom vanishing into vanishing of H^n(X, N) for n <= a.
- An object of D_QCoh on an affine scheme is determined by the complex of modules of its global derived sections. invoked at proof lines 10665-10667. Identifies the vanishing of cohomology with the vanishing of the truncation of N.

Source reference. Original remark 08CZ: sources/stacks-chow/tags.txt, source lines 10619–10619. Cited at statement line 10629 as the standing setup; the Koszul orthogonality argument is inside the counted span.

Source reference. Original lemma 062C: sources/stacks-chow/tags.txt, source lines 7791–7791. Cited at proof lines 10656-10657 to pass from K to the powers K_e; the Hom vanishing transfer is inside the counted span.

Source reference. Original lemma 08E3: sources/stacks-perfect/pinned-original-source.tex, source lines 2374–2374. Cited at proof lines 10659-10663 to convert Hom vanishing into cohomology vanishing; the truncation conclusion is inside the counted span.

Source reference. Original lemma 06Z0: sources/stacks-perfect/pinned-original-source.tex, source lines 454–454. Cited at proof lines 10666-10667 to identify cohomology vanishing with truncation vanishing; the final conclusion is inside the counted span.

Date: stacks-project:perfect.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.2. *In Situation 08CZ denote $j : U \to X$ the open immersion and let K be the perfect object of $D(\mathcal{O}_X)$ corresponding to the Koszul complex on $f_1, \ldots, f_r$ over A. Let $E \in D_{QCoh}(\mathcal{O}_X)$ and $a \in \mathbf{Z}$. Consider the following conditions*

(1) *The canonical map $\tau_{\leq a}E \to \tau_{\leq a}Rj_*(E|_U)$ is an isomorphism, and*
(2) $\mathrm{Hom}_{D(\mathcal{O}_X)}(K[-n], E) = 0$ *for all $n \leq a$.*

Then (2) implies (1) and (1) implies (2) with a replaced by $a - 1$.

Proof. Choose a distinguished triangle $E \to Rj_*(E|_U) \to N \to E[1]$. Then (1) implies $\tau_{\leq a-1}N = 0$ and (1) is implied by $\tau_{\leq a}N = 0$. Observe that

$$\mathrm{Hom}_{D(\mathcal{O}_X)}(K[-n], Rj_*(E|_U)) = \mathrm{Hom}_{D(\mathcal{O}_U)}(K|_U[-n], E) = 0$$

for all n as $K|_U = 0$. Thus (2) is equivalent to $\mathrm{Hom}_{D(\mathcal{O}_X)}(K[-n], N) = 0$ for all $n \leq a$. Observe that there are distinguished triangles

$$K^\bullet(f_1^{e_1}, \ldots, f_i^{e_i'}, \ldots, f_r^{e_r}) \to K^\bullet(f_1^{e_1}, \ldots, f_i^{e_i'+e_i''}, \ldots, f_r^{e_r}) \to K^\bullet(f_1^{e_1}, \ldots, f_i^{e_i''}, \ldots, f_r^{e_r}) \to \ldots$$

of Koszul complexes, see More on Algebra, Lemma 062C. Hence $\mathrm{Hom}_{D(\mathcal{O}_X)}(K[-n], N) = 0$ for all $n \leq a$ is equivalent to $\mathrm{Hom}_{D(\mathcal{O}_X)}(K_e[-n], N) = 0$ for all $n \leq a$ and all $e \geq 1$ with K_e as in Lemma 08E3. Since $N|_U = 0$, that lemma implies that this in turn is equivalent to $H^n(X, N) = 0$ for $n \leq a$. We conclude that (2) is equivalent to $\tau_{\leq a}N = 0$ since N is determined by the complex of A-modules $R\Gamma(X, N)$, see Lemma 06Z0. Thus we find that our lemma is true. □